

## **Assignment 1 - Firewalls**

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## **Introduction to Firewalls**

A firewall is a security device or software that monitors, filters, and controls inbound or outbound network traffic based on the security rules set on the firewall. The firewall inspects data packets which include source and destination addresses, port numbers, and protocol to determine if that type of network traffic should be permitted or blocked. The purpose of firewalls in a network is to make a barrier between your own trusted network and external networks that are not trusted, such as the networks on the internet. Firewalls prevent unauthorized access to your network and protects against malicious activities, preventing attacks and breaches of sensitive information. Configuration on a firewall can determine which inspected data packets are allowed or blocked since firewalls are gatekeepers between trusted and untrusted networks.

Modern firewalls are subject to changing technology yet include a wide range of advanced technological features and identify-based access controls. These advanced features are an important component in a network because it provides layers of defense while supporting zero trust principles and security compliance. You can make your network traffic strict with firewalls through access permissions while preventing the risk of security breaches on your network and ensuring confidentiality, integrity, and availability.

Digital Equipment Corporation developed packet filtering systems that was introduced in the late 1980s. IP addresses and port numbers are headers in a packet, and simple rules were set to allow or block traffic in first-generation firewalls. Stateful firewalls was a concept introduced in the early 1990s that can be credited to AT&T Bell Labs and Check Point Technologies. Stateful firewalls could track and manage the state of active network connections for security monitoring and real-time adaptability. Analysis of actual data contained in packets was introduced in the mid-1990s, where deeper inspection abilities was introduced in application-layer firewalls. Firewalls at this age could not only allow or block traffic based on packet headers, but defense against more sophisticated application-based attacks.

In the 2000s and beyond, firewalls were developed to have many capabilities bundled together, such as deep packet inspection, intrusion detection and prevent systems, and virtual private network support for remote access, network segmentation. Performance and flexibility problems that existed in earlier generations of firewalls were also addressed by adding granular security enforcement and application-aware controls. To this day, current technology such as artificial intelligence are used for threat detection to provide greater depth in defense for every layer in an organizational network.

## **Types of Firewalls**

### **Different Types of Firewalls**

There are many types of firewalls that provide methods for inspecting and controlling network traffic that can be customized for needs of security and the environment of the network. Packet-filtering firewalls and stateful inspection firewalls are two common types of firewalls. Firewalls that filter packets examine the IP address, port numbers, and protocols of every individual packet to allow or block the network traffic based on the configured rules. Stateful inspection firewalls track the state of active connections on top of checking the packet headers to filter traffic with greater detail and better monitor network sessions. Circuit-level gateways validate the connection, such as the process of a TCP handshake, but does not inspect the data. There are also web application firewalls that prevent SQL injection and cross-site scripting attacks by analyzing HTTP traffic. More comprehensive and multi-layered security features are offered by Modern Next-Generation Firewalls. These types of firewalls can have deep packet inspection, intrusion prevention, identify management, and threat intelligence integration features.

There are firewalls that protect the whole network and firewalls that are host-based. Network firewalls are at the entry or exit point between your internal network and the external network. Host-based firewalls are installed on individual devices such as your personal computer. We can also categorize these firewalls based on placement and scope in the network. Perimeter firewalls are placed at the edge of the network, internal firewalls are placed between different segments in the network, and distributed firewalls are placed on many endpoints in the network. Lastly, we have firewalls that are applications installed on computers or servers that protected against threats. Hardware firewalls are physical appliances connected to the network while software firewalls are programs on a device that are ideal for individuals or personal setups. Hardware firewalls are more commonly used in larger organizations rather than small groups or individuals.

### **Advantages and Disadvantages of Each Type of Firewall**

Packet-filtering firewalls are fast and cheap and provide simple traffic filtering for a network, but do not have many configuration options and are susceptible to more sophisticated network attacks. Stateful inspection firewalls provide better protection through state tracking of connections, but they require more system resources and continuous updates. Application-level firewalls provide a deep inspection at the operating system interface application layer, yet it can reduce data throughput and hinder compatibility with other protocols or applications. Next-generation firewalls involve many advanced features that provide comprehensive defense, but those firewalls cost more and have more demand for configuration and management.

Network firewalls can protect all devices on a network when installed and configured correctly, but a network firewall with security weakness or vulnerability can compromise all devices on a network. Host-based firewalls can better protect individual endpoints and be customized and configured for each device but can be expensive to install and implement in a large organization environment. The same applies to firewalls either placed at the perimeter of a network or firewalls that are distributed to many devices on a network.

## **Real-World Example**

The Finland-based VR Group manages and operates a high-speed train network. VR Group had the problem of slow internet speeds across trains, 5G integrations, low bandwidth, and lack of a network management system. VR Group worked with Forcepoint, a cybersecurity company, to install a next-generation firewall, create a high-speed network with remote access for IT staff for each train. VR can now connect with multiple mobile operators at the same time with high bandwidth and constant uptime with their new next-generation firewall. Now, customers experience enhanced connection capacity and staff can benefit from company-wide encryption for internal operations.

## **Firewall Functionality and Features**

### **Firewall Core Functionalities and Common Features**

One of the main functionalities of a firewall is to allow traffic from a trusted zone to an untrusted zone but not allow traffic from an untrusted zone to a trusted zone. Access control lists are policies that allow administrators to configure rules that determine which type of network traffic is permitted or denied. The control list regulates who has access to the network to prevent unauthorized users and potential threats. The network address translation firewall feature helps secure traffic by hides internal IP addresses behind a single public IP address. NAT allows devices on private networks to access the internet and block unsolicited inbound traffic at the same time. VPN is a service that creates an encrypted tunnel through the internet to connect a user to a private network as it hides their IP address and protects data in transit.

The following talking about other common features of firewalls. Protection of an endpoint involves installing the firewall in individual devices which provide an extra layer of security in the network by not only having a firewall at the perimeter of the network, but at the endpoints as well. Firewalls are also needed because of compliance and regulatory requirements. Many industries have certain security standards and compliances as part of their security policies which usually involve the best security policies. Firewalls also segment networks by dividing them into separate zones with each having their own custom configurations. Lastly, firewalls simply protect sensitive data and resources by denying threats of access and keeps your personal details or financial data confidential.

### **Feature Contribution to Network Security**

Apart from simple access controls, the feature of deny and allowing traffic of a firewall contributes to the security of a network because it acts as a gatekeeper that controls data based on rules for the security policy of an organization. This feature prevents attacks from gaining access to a network and prevents the spread of malware, viruses, and general intrusions. This feature also ensures that legitimate network activity occurs by validating the context of network packets. Firewalls these days also have controls with great levels of details which can further tailor to business needs and their specific services or applications.

## **Conclusion**

### **Key Takeaways and Their Role in Cybersecurity**

Key takeaways for firewalls all contribute to the security of networks and to prevent cyber threats. Firewalls monitor and filter network traffic to mainly block malicious data while preventing virus or malware. Firewalls protect sensitive information, provide data security, and enforce security policies. Firewalls also segment networks and reduce the impact of security breaches. Modern firewalls provide advanced threat protection through features such as deep packet inspection and virtual private networks. In general, firewalls enhance security by following the zero-trust rule and providing continuous validation of network traffic.

### **Importance of Firewalls in Network Protection from Cyber Threats**

Firewalls are usually the first line of defense when blocking hackers and cyber attackers, helping to maintain a secure environment from the internet. Users can make sure that they are working with only legitimate data when using firewalls as they use security policies to filter traffic. Firewalls also reduce the attack surface by segmentation of networks, dividing parts of the network to mitigate lateral movement of threats within the environment in the event of a breach. Overall, firewalls establish control and security in a network against evolving cyber threats.

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